

Optimised battery placement in distribution grids using quantum and quantum inspired QUBO solvers

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Abstract—The optimization of battery placement in distribution grids is a key challenge in the transition toward more reliable and resilient energy systems. This study evaluates the application of quantum and quantum-inspired solvers for addressing this problem in its Quadratic Unconstrained Binary Optimization (QUBO) formulations. The performance of various solvers, both quantum and quantum-inspired, is compared against traditional genetic algorithms. The analysis spans multiple test networks, including a large-scale real-world case study of the medium voltage grid in the Gipuzkoa region. Results indicate that quantum and quantum-inspired approaches can provide competitive or superior solutions balancing the three areas studied: investment costs, network reliability, and voltage control. Computational performance evaluations demonstrate that while simulated annealing remains the fastest among the tested approaches, hybrid quantum annealing successfully navigates complex optimization landscapes, reinforcing the feasibility of quantum computing in power grid applications.

Index Terms—Smart grids, distributed generation, QUBO, quantum annealing, quantum computation.

I. INTRODUCTION

A. Motivation

The transition to renewable energy sources within power grids is crucial for the mid-century targets of reduction and eventual elimination of CO₂ emissions [1][2][3]. Distributed energy resources (DERs) such as solar and wind power differ from traditional centralized sources (e.g., carbon, gas, hydroelectric, nuclear) in their geographic distribution, as DERs are usually smaller scale and located near consumption points. This introduces increased complexity in grid operation, as it poses many challenges regarding grid stability and power quality, for example managing bi-directional power flows and coordinating protection systems. Quantum computing and quantum-inspired solutions are proposed as valuable resources to address these challenges [4].

The primary objective of this project is to optimize battery placement as well as dimensioning across power grids to enhance voltage stability and robustness against failures

while minimizing installation costs. This task can be naturally formulated as a graph problem, with nodes representing sources, accumulators, and consumers connected by edges representing power lines. Such problems can be translated into QUBO problems, which are well-suited for quantum annealers [5], [6]. Quantum annealers, though more limited in scope than gate-based quantum computers, are better suited for binary optimization problems and can address larger instances. Quantum Approximation Optimization Algorithms (QAOA) also provide a solution framework for gate-based quantum computers [7].

B. State of the Art

Combinatorial optimization problems relevant to power grids can be expressed as QUBOs, including the maximum cut problem [8], minimal spanning tree [9]–[11], and quadratic assignment problem [6]. Quantum annealing has demonstrated effectiveness in solving these problems efficiently. For instance, community detection in electrical grids has shown that quantum and hybrid solvers outperform classical ones as problem sizes increase [12], [13]. Resource allocation problems, such as sensor placement in water networks [14]–[16] and electric vehicle charger placement [17], also benefit from quantum annealing's efficiency.

Quantum computing applications in power grids include power flow analysis using the Harrow-Hassidim-Lloyd (HHL) algorithm [18], [19], grid health monitoring [20], [21], true system-state estimation [22], resource coordination [5], [23]–[25], facility location-allocation [26], and minimizing ohmic losses through spanning tree optimization [27]. These applications demonstrate significant improvements in efficiency and performance over classical methods, especially when using quantum annealers and gate-based quantum computers.

Quantum-inspired algorithms, utilizing the mathematical principles of quantum mechanics, offer an alternative approach to optimization problems. Tensor Networks (TN),

implemented on conventional hardware like CPUs and GPUs, can solve graph problems and constrained combinatorial optimization problems. For example, QAOA-inspired TN solutions have shown efficiency in solving max-cut graph problems [28], [29], while quantum-inspired algorithms for constrained combinatorial optimization problems have demonstrated promising results [30]. TNs can represent any optimization problem expressible as a Hamiltonian, including higher-order unconstrained binary optimization (HUBO) problems [9].

In summary, both quantum and quantum-inspired solutions offer valuable tools for optimizing power grid operations and addressing key challenges in resource allocation and network management. Quantum annealers are particularly effective for large-scale optimization problems, while quantum-inspired algorithms provide a practical alternative using existing classical hardware. The advancements in these technologies present significant opportunities for improving the efficiency, reliability, and resilience of power grids in the presence of distributed energy resources.

C. Previous Works

The work presented in this project can be considered a continuation of the efforts carried out in the frame of the Flexener consortium¹, in which the same problem of battery allocation in a grid was tackled using a genetic algorithm (or GA) [31]. The goal of the present study is to explore quantum and quantum-inspired algorithms to improve the quality, particularly for larger networks, where the genetic algorithm seems to struggle. Although computational speed is a secondary factor—given that battery placement does not require frequent recalculations—it has also been considered during the evaluation phase. This allows us to assess whether quantum technologies can offer a computational advantage as problem complexity scales exponentially.

II. PROBLEM FORMULATION

A. QUBO Formulation

To find an optimal solution using a quantum annealer, it is essential to formulate the optimization problem in a QUBO (Quadratic Unconstrained Binary Optimization) form. This involves defining a set of binary variables $x_i \in \{0, 1\}$ and cost coefficients $q_{ij} \in \mathbb{R}$, such that the cost function is expressed as

$$C(\bar{x}) = \sum_{ij} q_{ij} x_i x_j = \bar{x}^T Q \bar{x},$$

where $\bar{x}^T$ denotes the transpose of the column vector $\bar{x}$, and Q represents the QUBO matrix. The diagonal elements of the QUBO matrix represent the linear part of the problem since for binary variables $x_i x_i = x_i$, while the off-diagonal elements capture the quadratic correlation costs. The goal of the QUBO problem is to find the solution vector that minimizes the cost function.

It is important to note that QUBO problems inherently do not support hard constraints. However, constraints can be introduced using additional terms in the cost function, often

involving extra binary variables known as slack variables. These types of constraints are referred to as soft constraints, as there is no systematic way to enforce them other than balancing them with the primary cost function.

To write this problem as a QUBO, the installed capacity per node χ_i is written in the form of an integer variable (easily translated to binary), by setting a unit cell capacity c and considering any installed capacity the sum of y_i of these unit cells

$$\chi_i = c \cdot y_i.$$

The cost function that defines the QUBO matrix Q will be defined in the following section

B. Cost Function

According to [31], the problem is framed as a multi-objective optimization involving voltage control under normal operation, reliability improvement, and monetary investment when installing batteries into the grid.

a) *Reliability*: The network is divided into zones² delimited by switches intended for isolating certain areas in case of fault. Within each zone, battery location does not impact overall reliability. The reliability metric, which we seek to minimize, is the economic impact of a failure:

$$C_R = \sum_i^N \sum_j^N f_i \tau_i \epsilon_j Z_{i,j},$$

where f_i is the failure rate of zone i , τ_i is the time needed to repair zone i after a fault, ϵ_j is the economic impact of an affected zone j losing supply, and $Z_{i,j}$ is the interaction matrix between zones that gives us the fraction of the demand not being satisfied.

When a failure occurs in zone i , we can divide the grid into microgrids that are no longer connected. If we define S_j^i as the set of nodes still connected to j when a failure occurs in i , the interaction matrix can be defined as:

$$Z_{i,j}^{\text{int}} = \begin{cases} 0 & j \text{ upstream} \\ 1 & i = j \\ \max \left\{ 0, 1 - \frac{\sum_{n \in S_j^i} (\chi_n + g_n \tau_i)}{\sum_{n \in S_j^i} d_n \tau_i} \right\} & j \text{ downstream} \end{cases},$$

where χ_n , g_n , and d_n are the capacity, the average generated power, and the average power demand at node n , downstream of node i .

The previous equation defines the reliability as a piece-wise function, where, as we increase the installed capacity and the power demands of the different zones are met, the returns in reliability are smaller, until reaching a saturation point (C_{sat}) beyond which the reliability will no longer increase, when the power demand of the zone has been completely satisfied (R_{max}). In order to write it into a QUBO, we take the quadratic approximation as shown in figure 1, setting the vertex of the parabola on the saturation point of the

²Since we will be working with a graph representation of this network the terms *zone* and *node* will be used interchangeably through this paper to designate the group of buses delimited by these switches.

¹Flexener project

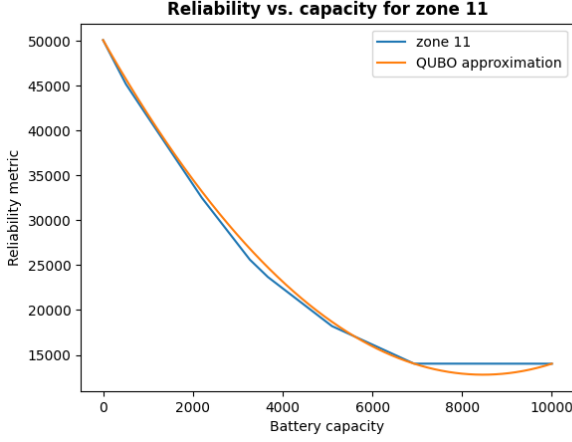


Fig. 1. Behavior of the reliability metric when changing the capacity installed in zone 11 of an example network (blue), and the quadratic approximation used to build the QUBO formulation (orange)

original function. In a maximization/minimization problem, we ignore terms independent of χ_i and maximize instead of minimizing, focusing on maximizing monetary savings from battery installation in case of failure. Applying these changes, the reliability expression

$$1 - \frac{\sum_j \chi_j + g_j \tau_j}{\sum_j d_j \tau_j}$$

becomes a quadratic function passing through (0, 0):

$$C_R = \sum_i a_i \chi_i^2 + b_i \chi_i. \quad (1)$$

To fit the parabola through maximum reliability (R_{max}) and saturation capacity (C_{sat}), parameters are set as:

$$\begin{cases} a = \frac{-R_{max}}{C_{max} C_{sat}} \\ b = -a(C_{sat} + C_{max}) \end{cases}. \quad (2)$$

For a single battery type, reliability (2) in zone i becomes:

$$C_i^{rel} = a_i \chi_i^2 + b_i \chi_i. \quad (3)$$

A penalty is needed for battery distribution across zones, to more accurately model the interaction between nearby nodes. We introduce a penalty $-k x_i x_j$ when zones i and j are in the same microgrid:

$$-\frac{R_{max}}{P_d^2} x_i x_j. \quad (4)$$

This way, if a battery is installed in a zone j sharing the microgrid with i , a penalty is incurred, since installing batteries in the same microgrid does not improve the reliability much.

b) Voltage Control: For each zone, the node providing the best voltage control improvement per installed capacity is selected. If V_n denotes the voltage control value for zone n , obtained by calculating the variation in voltage associated with a variation in power injections in each zone through a power-flow analysis [31], the voltage control metric, which we seek to maximize, is:

$$C_{VC} = \sum_n^N V_n \cdot \chi_n.$$

c) Investment: The investment metric can be approximated using either a linear or quadratic form. The quadratic approximation provides a more realistic representation by accounting for economies of scale, and is expressed as:

$$C_I = \sum_n^N a \chi_n^2 + b \chi_n,$$

where b depends on the unit cost of batteries and a is a correction factor accounting for economies of scale. This approximation, based on real battery cost data by i-DE (the Spanish electricity distribution company within the Iberdrola group), and other external reports [32] is valid for capacities under 10,000 kWh.

C. Comparing solutions in multi-objective optimization tasks

The problem addressed in this study falls within the domain of multi-objective optimization, where several objective functions must be considered simultaneously. In such contexts, the optimal solution is not unique but rather represented by a set of non-dominated solutions known as the Pareto front, which illustrates the trade-offs between competing objectives. From this, a final solution is typically selected *a posteriori* based on the decision-maker's priorities. Our primary benchmark is a previously developed genetic algorithm [31], which generates this Pareto front. Each solution on the front represents a different balance among objectives such as system reliability, voltage control, and investment cost. Some of these objectives are inherently conflicting—for instance, improving reliability and voltage control typically requires higher installed capacity, while minimizing investment tends to reduce it. The genetic algorithm enables exploration of the solution space without imposing *a priori* preferences.

On the other hand, in methods like QUBO, in which we seek a single solution, this selection must be made *a priori*. There are several strategies for adapting a multi-objective problem to a single-objective solver [33]. Common approaches include: (i) transforming some objectives into constraints (e.g., by imposing upper bounds), or (ii) combining all objectives into a single composite function. Given that QUBO solvers struggle with hard constraint satisfaction—treating them as soft penalties that complicate the cost function—we opted for the second approach. Specifically, we employed a weighted linear combination of the objective functions, where each term C_α is multiplied by a coefficient (or Lagrange multiplier) λ_α representing its relative weight. This formulation allows for explicit control over the importance of each criterion, facilitating both interpretation and implementation in quantum or quantum-inspired architectures.

$$C(\bar{x}) = \sum_\alpha \lambda_\alpha C_\alpha(\bar{x}),$$

where α takes the values in the set $\alpha \in \{R, VC, I\}$ for the reliability, voltage control, and investment terms, respectively.

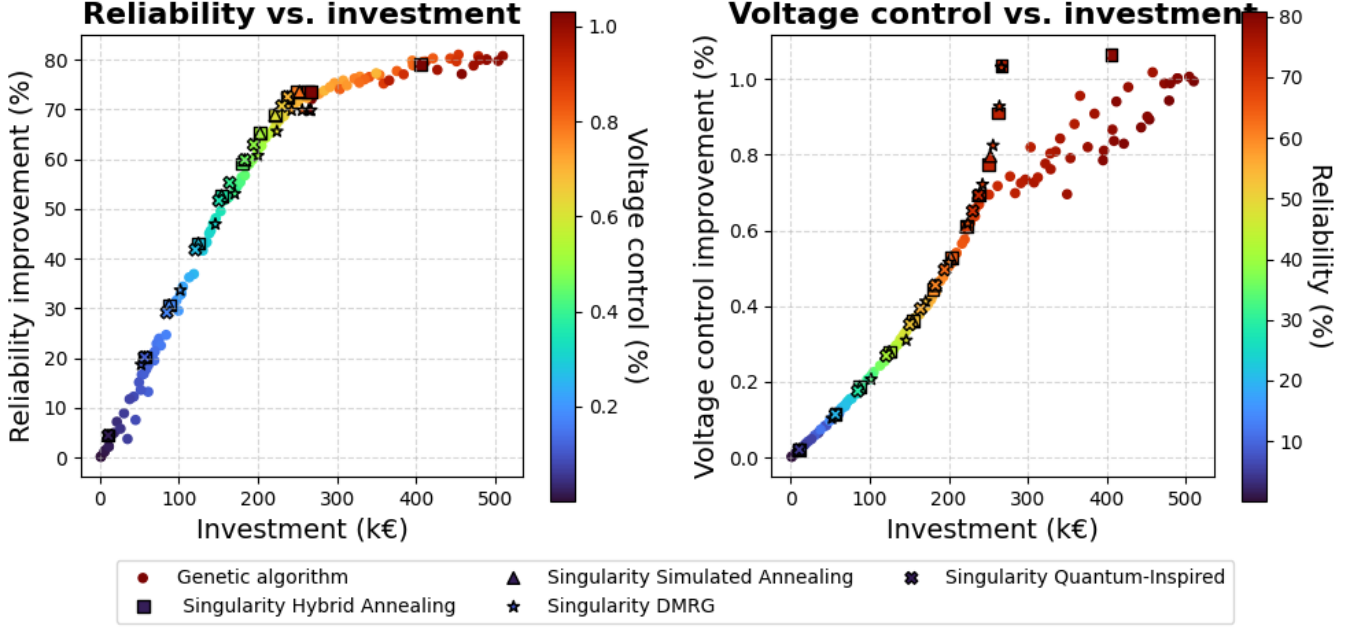


Fig. 2. Comparison between the three metrics of the solutions generated by the genetic algorithm (dots) and QUBO solutions obtained using different backends available in Singularity.

To make sure all terms affect the optimization problem, we first approximate the order of magnitude by installing the maximum allowed capacity and set the weights accordingly, so no term dominates while others are ignored during the optimization.

After obtaining this single-objective, single-solution problem, to enable a meaningful comparison with our benchmark, we adopted a systematic strategy: we varied the maximum installed capacity constraint. This approach allows us to generate solutions that, while not forming an explicit Pareto front, capture different points in the trade-off space. Each point in Figure 2 corresponds to a solution obtained under a different setting of this constraint, enabling us to assess system behavior under varying investment levels. Although adjusting the relative weights of the objective function terms is a common alternative, it introduces greater uncertainty and less control over the resulting solutions. In contrast, modifying the maximum installed capacity constraint provides a more direct and reproducible mechanism for exploring the solution space and validating the effectiveness of the proposed weights. For larger grid instances, experts from Iberdrola evaluated the quality of the solutions by verifying whether the prioritized nodes aligned with the grid’s topology and operational characteristics.

D. Quantum and Quantum-Inspired Solvers

For solving QUBO problems, we employ a suite of advanced techniques, all integrated within Singularity Optimization (SOP), a proprietary software platform developed by Multiverse Computing. Singularity offers easy and practical access to quantum and quantum-inspired optimization meth-

ods (backends), enabling users to efficiently tackle complex optimization problems.

For this project we have used the following backends:

- 1) **Simulated annealing** is a meta-heuristic optimization technique inspired by the annealing process in metallurgy, which uses random “thermal” jumps to explore the solution space and progressively smaller jumps as the “temperature” decreases to ensure the solution remains at a point of minimal energy. Multiple runs are executed to ensure a global minimum.
- 2) **Quantum hybrid annealing** builds on this by leveraging quantum properties such as superposition and quantum tunneling, just like quantum annealing, but using a hybrid approach (combination with classical techniques) to overcome the current limitations of quantum hardware, allowing to solve larger size problems and obtain less noisy results.
- 3) **Tensor network optimizers**, which use a mathematical framework from quantum mechanics to describe entangled quantum systems and employ algorithms like the Density Matrix Renormalization Group (DMRG) to efficiently approximate the ground state of the system on classical hardware, offering advantages in cybersecurity and data sensitivity.
- 4) **Multiverse quantum-inspired solver**, combines simulated annealing with tensor network truncation and compression methods to enhance solver performance, albeit with a slight loss in accuracy.

III. RESULTS

In this section, we present results from four different networks. We first examine three networks from the previous

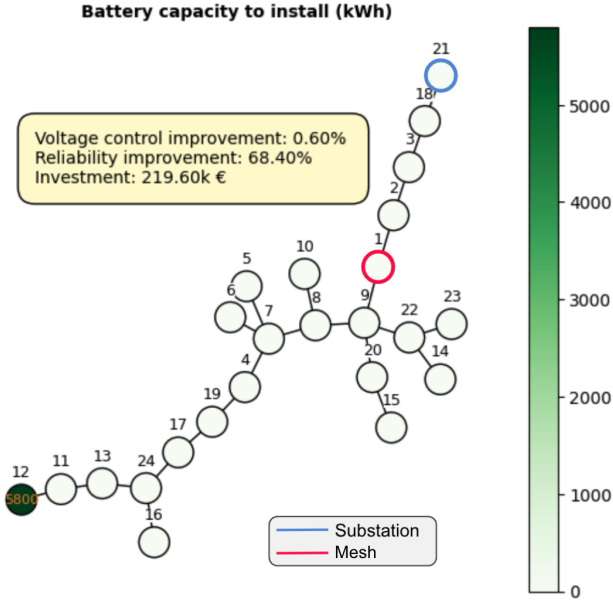


Fig. 3. Single weight set results using quantum hybrid annealing. In blue, the upstream node; in red, zones that can connect to other grids.

Flexener project [31] to compare with the genetic algorithm developed in that project and calibrate our algorithm. Then, we study a larger network (1968 zones - we recall the term “zone” is often used equivalently to the term “node” in graph theory) corresponding to the medium voltage network of Gipuzkoa province.

For the smaller grids, we focus on solution quality and proximity to GA results to validate our algorithm before applying it to the larger network. An exhaustive runtime study is performed only for the larger network to demonstrate the advantage of quantum computing.

We present results from four different Singularity backends as described in the previous section, all addressing the same QUBO formulation.

A. Toy Network

We compare our single solution obtained with the D-Wave hybrid solver against GA results. Figure 3 shows a solution using recommended weights and a maximum capacity of 10,000 kWh with 100 kWh battery size.

Figure 2 shows Pareto comparisons using different solvers.

B. Gipuzkoa Medium Voltage Network

One of the main objectives of the project is to examine the performance of these quantum and quantum-inspired solvers in larger or more complex cases. However, the pure quantum annealer could not handle the number of variables necessary for this large of a problem. For this reason, we used the quantum hybrid solver to study a use case derived from the medium voltage network of the Gipuzkoa province, with a size of 1968 nodes.

To run these results, we made a couple of changes. Firstly, the relevant metrics for the cost function were reliability and

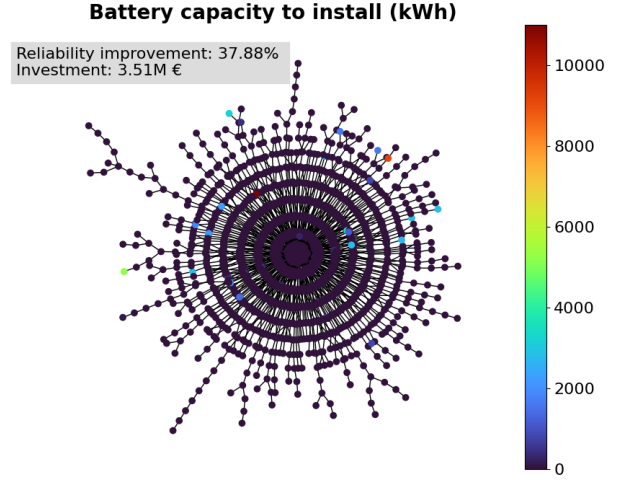


Fig. 4. Singularity hybrid annealing solution for the Gipuzkoa medium voltage network.

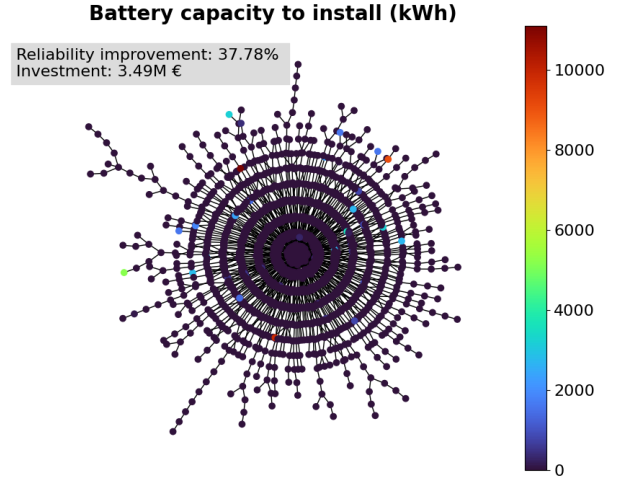


Fig. 5. Singularity simulated annealing solution for the Gipuzkoa medium voltage network.

investment due to the computational complexity of calculating the voltage control improvement. Secondly, we adjusted the quadratic expression for investment to allow for higher installation capacities and removed the maximum capacity constraint to reduce the number of terms in the QUBO matrix.

Figure 4 shows a solution for this network. In an analysis performed by Iberdrola, the algorithm prioritizes zones with high overhead line lengths, high Value of Lost Load (VOLL -i.e. the economical loss associated to interrupting the service at the given zone), and extremal nodes, aligning with expected behavior, according to the cost function.

Figure 5 shows similar solutions obtained using simulated annealing.

Figure 6 shows the comparison between our results and the GA results, demonstrating superior performance in reliability for similar investment values.

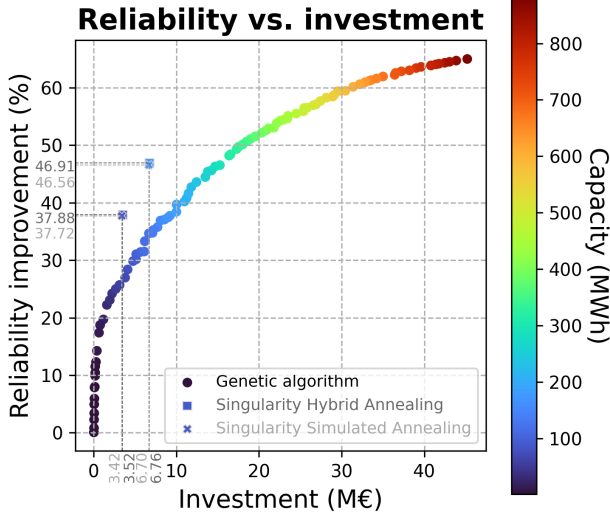


Fig. 6. Comparison between genetic algorithm solution and Singularity solutions.

C. Time Analysis

An in-depth study of the time taken by both simulated annealing and hybrid annealing solvers was performed. The analysis focuses on the optimization part, excluding preprocessing and postprocessing times. Table I presents the time distribution for different network sizes.

Figure 7 shows the runtimes of the genetic algorithm for networks of different sizes, demonstrating that both hybrid and simulated annealing solvers are faster.

IV. CONCLUSIONS AND OUTLOOK

This study has evaluated the performance of quantum and quantum-inspired solvers on a series of network optimization problems, offering a comprehensive comparison against traditional genetic algorithms. As a continuation of a previous study [31], we analyzed the same small sized networks, of which one of them is presented in this paper, and also extended to a significantly larger network representing the medium voltage network of the Gipuzkoa province. The latter case was particularly challenging, highlighting the limitations of genetic algorithms in handling complex, large-scale networks.

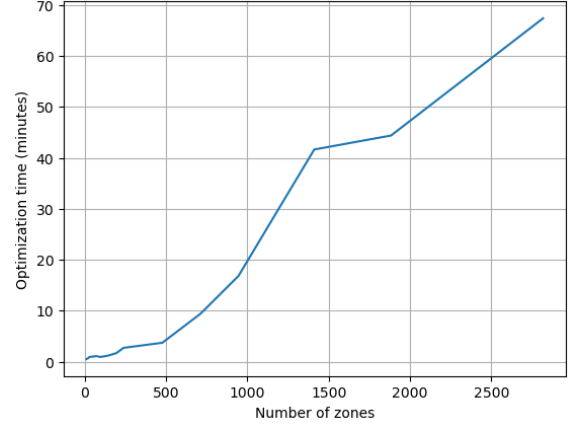


Fig. 7. Runtimes for the optimization of different sized networks, using the genetic algorithm

The quantum-inspired solvers, including simulated annealing, hybrid quantum annealing, a tensor network optimizer based in TEBD, and a second quantum-inspired solver which combines simulated annealing and tensor network, were applied to the same QUBO formulation. These solvers demonstrated varying degrees of success across the networks, showing promising results in terms of finding solutions that balance key metrics.

For the smaller network, comparisons with the genetic algorithm revealed that quantum and quantum-inspired approaches could indeed find competitive or superior solutions, particularly in scenarios requiring high reliability and specific investment constraints. Notably, the hybrid and simulated annealing approaches demonstrated significant potential in navigating the solution space effectively.

In the case of the Gipuzkoa network, we also report an improvement of the quality of the solutions. With the same investment we provide solutions that noticeably improve the reliability of the network. The computational time to obtain the results was also improved. This is due to multiple reasons: the first one being that instead of a Pareto front provided by the GA, our solution provides a single-point solution. This reduces the computational time as it focus the attention to a single result instead of providing a set of valid results. Furthermore, the GA solution places small capacity in all the nodes, reducing the quality of the solution as it significantly increases the investment. To extract a practical solution from the GA solution, it is first necessary to analyze the solution

TABLE I
COMPUTATION TIME DISTRIBUTION OF THE OPTIMIZATION PROCESS FOR DIFFERENT SOLVERS AND DIFFERENT NETWORK SIZES

		20 zones		1968 zones		4654 zones	
		SA	HA	SA	HA	SA	HA
Build Graph		8.99×10^{-4} s		2.88s		17.95s	
Write QUBO	Reliability	2.45×10^{-3} s		9.99s		65.93s	
	Investment	7.55×10^{-5} s		3.54×10^{-3} s		0.023s	
	Voltage control	1.71×10^{-4} s		-		-	
Singularity backend		2.60s	11.81s	17.99s	119.23s	118.25s	288.75s
Sampling (QPU time)		0.02s	≤ 3 s (0.21s)	4.59s	85.92s (0.51s)	3	214.60s (0.86s)
Total		2.60s	11.81s	33.74s	134.98s	220.10s	390.60s

and identify the zones in which most of the capacity is being placed. Whereas the solution obtained with the QUBO formulation provides a solution that does not require further analysis, installing capacity only in strategic nodes.

We notice that simulated annealing still proves to be faster in the studied cases w.r.t. hybrid annealing, pointing to the conclusion that this is still a classically solvable problem. Nonetheless we demonstrate that it is possible to solve this problem on a quantum annealer, proving a successful application of quantum computing to the energy networks sector. This allows us to envision the possibility of solving larger and more complex use cases in the future. Specially with the advancement in quantum hardware, that has been rapidly improving in the latest years promising further advancements in the next future.

Overall, our findings highlight the viability and potential advantages of quantum and quantum-inspired solvers for network optimization tasks, especially as these technologies continue to mature, even though it is important to note that, with the current state of quantum technologies, we have only proved that these advantages can be leveraged for combinatorial optimization problems like the one studied, where quantum annealers can show their maximum potential, and not any kind of problem. While traditional methods like genetic algorithms have their place, the scalability and efficiency of quantum and quantum-inspired solutions offer a compelling alternative for tackling the complexities inherent in large-scale, real-world networks, where complexity does not grow linearly but exponentially, as each additional zone or node increases the number of interconnections within the grid. Further research and development in this field will only lead to more robust, efficient, and scalable solutions.

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